

Packet Tracer - Identify MAC and IP Addresses

Objectives

Part 1: Gather PDU Information for Local Network Communication

Part 2: Gather PDU Information for Remote Network Communication

Background

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

Instructions

Part 1: Gather PDU Information for Local Network Communication

Note: Review the Reflection Questions in Part 3 before proceeding with Part 1. It will give you an idea of the type of information you will need to gather. Gather PDU information as a packet travels from 172.16.31.5 to 172.16.31.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 172.16.31.2** command.
- Switch to simulation mode and repeat the **ping 172.16.31.2** command. A PDU appears next to **172.16.31.5**.
- Click the PDU and note the following information from the **OSI Model** and **Outbound PDU Layer** tabs:
 - Destination MAC Address: **000C:85CC:1DA7**
 - Source MAC Address: **00D0:D311:C788**
 - Source IP Address: **172.16.31.5**
 - Destination IP Address: **172.16.31.2**
 - At Device: **172.16.31.5**
- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered into a spreadsheet using a format like the table shown below:

Example Spreadsheet Format

At Device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4	Type
172.16.31.5	000C:85CC:1DA7	00D0:D311:C788	172.16.31.5	172.16.31.2	Ethernet
Switch1	000C:85CC:1DA7	00D0:D311:C788	172.16.31.5	172.16.31.2	Ethernet
Hub	000C:85CC:1DA7	00D0:D311:C788	172.16.31.5	172.16.31.2	Ethernet
172.16.31.2	00D0:D311:C788	000C:85CC:1DA7	172.16.31.2	172.16.31.5	Ethernet

At Device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4	Type
172.16.31.2	N/A	N/A	172.16.31.2	172.16.31.3	IP
172.16.31.2	0060.7036.2849	000C:85CC:1DA7	172.16.31.2	172.16.31.3	Ethernet
Hub	0060.7036.2849	000C:85CC:1DA7	172.16.31.2	172.16.31.3	Ethernet
Switch1-x	0060.7036.2849	000C:85CC:1DA7	172.16.31.2	172.16.31.3	Ethernet
172.16.31.3	000C:85CC:1DA7	0060.7036.2849	172.16.31.3	172.16.31.2	Ethernet
172.16.31.4	N/A	N/A	172.16.31.4	172.16.31.5	IP
172.16.31.4	00D0.D311.C788	000C.CF0B.BC80	172.16.31.4	172.16.31.5	Ethernet
Switch1	00D0.D311.C788	000C.CF0B.BC80	172.16.31.4	172.16.31.5	Ethernet
172.16.31.5	000C.CF0B.BC80	00D0.D311.C788	172.16.31.5	172.16.31.4	Ethernet

Step 2: Gather additional PDU information from other pings.

Repeat the process in Step 1 and gather the information for the following tests:

· Ping 172.16.31.2 from 172.16.31.3

· Ping 172.16.31.4 from 172.16.31.5

Return to Realtime mode.

Part 2: Gather PDU Information for Remote Network Communication

In order to communicate with remote networks, a gateway device is necessary. Study the process that takes place to communicate with devices on the remote network. Pay close attention to the MAC addresses used.

Step 1: Gather PDU information as a packet travels from 172.16.31.5 to 10.10.10.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 10.10.10.2** command.
- Switch to simulation mode and repeat the **ping 10.10.10.2** command. A PDU appears next to **172.16.31.5**.
- Click the PDU and note the following information from the **Outbound PDU Layer** tab:
 - Destination MAC Address: 00D0:BA8E:741A, Source MAC Address: 00D0:D311:C788, Source IP Address: 172.16.31.5, Destination IP Address: 10.10.10.2, At Device: 172.16.31.5

What device has the destination MAC that is shown?

- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered from pinging 172.16.31.5 to 10.10.10.2 into a spreadsheet using a format like the sample table shown below:

PDU Data Table

At Device	Dest. MAC 1 Ethernet Header	MAC 2 Address	Src MAC 1 Ethernet Header	Src IPv4 IP Header	Dest IPv4 IP Header	Packet Direction	Type	Highest OSI Layer
172.16.31.5	000C.85CC.1DA7		00D0:D311:C788	172.16.31.5	172.16.31.2	outbound	Gigabit Ethernet	3 network
Switch1	000C.85CC.1DA7		00D0:D311:C788	172.16.31.5	172.16.31.2	Outbound	Gigabit Ethernet	2 data link
Hub	000C.85CC.1DA7		00D0:D311:C788	172.16.31.5	172.16.31.2	Outbound	Fast Ethernet	1 physical
172.16.31.2	00D0:D311:C788		000C.85CC.1DA7	172.16.31.2	172.16.31.5	Outbound	Gigabit Ethernet	3 network
172.16.31.3	000C.85CC.1DA7		00D0:D311:C788	172.16.31.5	172.16.31.2	Inbound	Gigabit Ethernet	0
Hub	00D0:D311:C788		000C.85CC.1DA7	172.16.31.2	172.16.31.5	Outbound, ACK	Fast Ethernet	1 physical
Switch1	00D0:D311:C788		000C.85CC.1DA7	172.16.31.2	172.16.31.5	Outbound, ACK	Gigabit Ethernet	2 data link
172.16.31.3	00D0:D311:C788		000C.85CC.1DA7	172.16.31.2	172.16.31.5	Inbound	Gigabit Ethernet	2 data link
172.16.31.5	00D0:D311:C788		000C.85CC.1DA7	172.16.31.2	172.16.31.5	Inbound,A CK	Gigabit Ethernet	3 network
172.16.31.3	00D0.BA8E.741A		0060.7036.2849	172.16.31.3	10.10.10.3	outbound	Gigabit Ethernet	3 network

At Device	Dest. MAC 1 Ethernet Header	MAC 2 Address	Src MAC 1 Ethernet Header	Src IPv4 IP Header	Dest IPv4 IP Header	Packet Direction	Type	Highest OSI Layer
Hub	00D0.BA8E.741A		0060.7036.2849	172.16.31.3	10.10.10.3	Outbound	Fast Ethernet	1 physical
Switch1	00D0.BA8E.741A		0060.7036.2849	172.16.31.3	10.10.10.3	Outbound	Gigabit Ethernet	2 data link
172.16.31.2	00D0.BA8E.741A		0060.7036.2849	172.16.31.3	10.10.10.3	Inbound	Gigabit Ethernet	2 data link
Router	00D0.BA8E.741A		0060.7036.2849	172.16.31.3	10.10.10.3	Outbound	Gigabit Ethernet	3 network
Router	0060.4706.572B		00D0.588C.2401	172.16.31.3	10.10.10.3	Outbound	Gigabit Ethernet	3 network
Switch0	0060.4706.572B		00D0.588C.2401	172.16.31.3	10.10.10.3	Outbound	Gigabit Ethernet	2 data link
Access Point	0060.4706.572B		00D0.588C.2401	172.16.31.3	10.10.10.3	Inbound	Gigabit Ethernet	1 physical
Access Point	0060.4706.572B	0050.0FAB.6C82	00D0.588C.2401	172.16.31.3	10.10.10.3	Outbound	802.11 Wireless	1 physical
10.10.10.3	0050.0FAB.6C82	0060.4706.572B	00D0.588C.2401	10.10.10.3	172.16.31.3	Outbound	802.11 Wireless	3 network
10.10.10.2	0060.4706.572B	0050.0FAB.6C82	00D0.588C.2401	172.16.31.3	10.10.10.3	Inbound	802.11 Wireless	1 physical
10.10.10.3	0050.0FAB.6C82	0060.4706.572B	00D0.588C.2401	10.10.10.3	172.16.31.3	Outbound	802.11 Wireless	1 physical

At Device	Dest. MAC 1 Ethernet Header	MAC 2 Address	Src MAC 1 Ethernet Header	Src IPv4 IP Header	Dest IPv4 IP Header	Packet Direction	Type	Highest OSI Layer
Access Point	00D0.588C.2401		0060.4706.572B	10.10.10.3	172.16.31.3	Outbound, ACK	Ethernet	1 physical
Switch0	00D0.588C.2401		0060.4706.572B	10.10.10.3	172.16.31.3	Outbound, ACK	Ethernet	2 data link
Router	0060.7036.2849		00D0.BA8E.741A	10.10.10.3	172.16.31.3	Outbound, ACK	Ethernet	3 network
Access Point	00D0.588C.2401	0050.0FAB.6C82	0060.4706.572B	10.10.10.3	172.16.31.3	Outbound	Ethernet	1 physical
10.10.10.3	00D0:588C:2401	0050.0FAB.6C82	0060.4706.572B	10.10.10.2	172.16.31.3	Inbound	802.11 Wireless	1 physical
10.10.10.2	0060.4706.572B	0050.0FAB.6C82	00D0.588C.2401	172.16.31.3	10.10.10.3	Inbound	802.11 Wireless	1 physical
Switch1	0060.7036.2849		00D0.BA8E.741A	10.10.10.3	172.16.31.3	Outbound, ACK	Gigabit Ethernet	2 data link
Hub	0060.7036.2849		00D0.BA8E.741A	10.10.10.3	172.16.31.3	Outbound, ACK	Fast Ethernet	1 physical
172.16.31.2	0060.7036.2849		00D0.BA8E.741A	10.10.10.3	172.16.31.3	Inbound	Gigabit Ethernet	3 network
172.16.31.3	0060.7036.2849		00D0.BA8E.741A	10.10.10.3	172.16.31.3	Inbound, ACK	Gigabit Ethernet	3 network

Reflection Questions

Answer the following questions regarding the captured data:

1. Were there different types of cables/media used to connect devices? **Wired Ethernet and wireless**
2. Did the cables change the handling of the PDU in any way? **No. The transmission media (Ethernet cable or wireless) did not change the IP packet. Different media only changed the Layer 2 frame format used to carry the packet.**
3. Did the **Hub** lose any of the information that it received? **No, but it sent information to a device that dropped the PDU b/c the device was not the destination.**
4. What does the **Hub** do with MAC addresses and IP addresses? **A hub does not examine or modify MAC or IP addresses. It simply repeats incoming electrical signals out every other port.**
5. Did the wireless **Access Point** do anything with the information given to it? **No. the IP packet stays exactly the same.**
6. Was any MAC or IP address lost during the wireless transfer? **No**
7. What was the highest OSI layer that the **Hub** and **Access Point** used? **The Hub reached layer 1 physical and the AP reached layer 2 data link**
8. Did the **Hub** or **Access Point** ever replicate a PDU that was rejected with a red "X"? **No, there was never an ACK sent back from the node that rejected the PDU**
9. When examining the **PDU Details** tab, which MAC address appeared first, the source or the destination? **Both the AP and the HUB have the destination first, except when the AP has 4 MAC address spaces.**
10. Why would the MAC addresses appear in this order? **Because the purpose is delivery to the destination address so it's first**
11. Was there a pattern to the MAC addressing in the simulation? **Yes, when the PDU is traveling to the destination, then the destination is first, but once at the destination, the source and destination flip and the ACK is then sent to the initial source (now the destination).**
12. Did the switches ever replicate a PDU that was rejected with a red "X"? **No, there was never an ACK sent back from the node that rejected the PDU. The ACK that was sent from the target node was replicated to all connected devices.**
13. Every time that the PDU was sent between the 10 network and the 172 network, there was a point where the MAC addresses suddenly changed. Where did that occur? **It changed at the router, where the MAC address changes at every hop.**
14. Which device uses MAC addresses that start with 00D0:BA? **172.16.31.3**
15. What devices did the other MAC addresses belong to? **10.10.10.3**

16. Did the sending and receiving IPv4 addresses change fields in any of the PDUs? **Yes, when the packet was received by the target device, the ACK as the destination and source IP flipped.**
17. When you follow the reply to a ping, sometimes called a *pong*, do you see the sending and receiving IPv4 addresses switch? **Yes, the pong "ACK" is reversed from the initial request.**
18. What is the pattern to the IPv4 addressing used in this simulation? **Each port on the router belongs to one network. One port leads to the 172 thru switch 0 and the other leads to the 10.10 thru switch 1. The route connects together.**
19. Why do different IP networks need to be assigned to different ports of a router? **Each router interface belongs to a different IP network. Routers separate broadcast domains and provide routing between different IP subnets.**
20. If this simulation was configured with IPv6 instead of IPv4, what would be different? **The communication process would be the same, but IPv6 addresses would replace IPv4 addresses, ARP would be replaced by Neighbor Discovery (NDP), and ICMPv6 would be used instead of ICMP for address resolution and other functions.**